REPORT

Reviewer report: urllib3/urllib3#5278: HTTP/2 prototype

Run urllib3-5278, tested 2026-11-11 11:46 IST by ops/daily_run.py (route source-pair). Proof-of-Change commit

ef12096677e9cf59e8673436d74ec65d53771d7c. Report generated 2026-11-11 12:44 IST by

reports/pr_report.py.

Baseline: full (test comparison + claim probes + coverage gate + claim flagging (differential probes shown,

informational)).

NEEDS ATTENTION

Recommended action: Reject Confidence: High

The run is trustworthy and found a difference a person should look at. Every check passed and every claim cites

a log line.

Advice only: Proof-of-Change never blocks a merge and never posts on GitHub.

Summary in plain English

• What it says it does: HTTP/2 prototype (urllib3/urllib3#5278).

• What we observed: 10 new test(s) fail on the PR; 2 test(s) removed on the PR side; 839 test(s) ran on main,

865 on the PR; 18 new test(s) pass, 366 unchanged (26 of them skipped on both sides); 497 test failure(s)

pre-existing and identical on both sides.

• What you should do: Do not approve as is. Read 'What changed in behavior' and decide with the PR author.

Demo helper

These notes are background from the console and the daily check. They do not change the verdict.

If setup fails, the helper retries the install up to 2 times. If a test still can't be verified because something is

missing or fails to import, it looks up the cause, installs the fix on both sides and reruns that step, up to 3 tries.

• Setup check: GitHub token: present.

• Setup check: Nebius token: present.

• Setup check: free memory 11.5 GB (of 15.6 GB).

• Setup check: free disk 92.2 GB.

• Demo helper: git is installed.

• Demo helper: python3 is installed.

• Demo helper: uv is installed.

• Looked up with Tavily: [redacted] results

• Query: urllib3 python setup install instructions https://github.com/urllib3/urllib3 site:github.com/urllib3/urllib3

• Query: urllib3 python documentation install https://github.com/urllib3/urllib3 site:github.com/urllib3/urllib3

• Query: urllib3 python install requirements https://pypi.org/project/urllib3/ https://github.com/urllib3/urllib3

site:pypi.org

• Source: urllib3 · PyPI — https://pypi.org/project/urllib3/

• Source: urllib3 installation requires setuptools >= 36.2.5 · Issue #1753 · urllib3/urllib3 · GitHub —

https://github.com/urllib3/urllib3/issues/1753

• Source: update of urllib3 makes kodi start up much slower · Issue #1590 · urllib3/urllib3 · GitHub —

https://github.com/urllib3/urllib3/issues/1590

• Source: urllib3/docs/advanced-usage.rst at main · urllib3/urllib3 · GitHub —

https://github.com/urllib3/urllib3/blob/master/docs/advanced-usage.rst

• Source: GitHub - urllib3/urllib3: urllib3 is a user-friendly HTTP client library for Python · GitHub —

https://github.com/urllib3/urllib3

• Source: Installing urllib3 · Issue #829 · urllib3/urllib3 · GitHub — https://github.com/urllib3/urllib3/issues/829

• Source: urllib3>=2.0 does not work with system Python on macOS —

https://github.com/urllib3/urllib3/issues/3020?timeline_page=1

• Setup plan: the same packages on main and pr.

• Setup plan: install trustme>=1.2.1 (listed in the test extra) on main and pr.

• Setup plan: Python 3.12: Python 3.12 already meets >=3.10

• Installed trustme 1.2.1 from pyproject (listed in the test extra) on main and pr.

• Before rerun: needs_attention: 10 new test(s) fail on the PR; 2 test(s) removed on the PR side; 839 test(s) ran

on main, 865 on the PR; 18 new test(s) pass, 366 unchanged (26 of them skipped on both sides); 497 test

failure(s) pre-existing and identical on both sides

• After rerun: not rerun. The collection import was a setup gap, but it was not repaired, so the recorded verdict

was kept.

Setup: pytest plugin asyncio was turned off. Pytest stopped at startup the same way on both sides, and the

traceback pointed into that third-party plugin (AttributeError: module 'pytest' has no attribute

'pyodide_options_stack'). The tests were run again with the plugin turned off. This does not change how the

verdict is chosen. Detection fired. Pytest stopped at startup the same way on both sides. Plugin chosen: asyncio.

The installed pytest plugin list was read. Retry flags: -p no:asyncio. Outcome: worked.

Claims vs reality

| # | Claim (source) | Status | What we observed | Log line

| C1 | HTTP/2 prototype (PR title) | not testable | claims are read out of the PR description only by model-written

probes (--model-provider, see Probes); the tests below check behavior | pytest-pr-1.log:51413 «130 failed, 358

passed, 18 skipped, 8 xfailed, 145 warnings, 507 errors in 173.93s (0:02:53)»

| C2 | Existing behavior is kept: tests that pass on main still pass (implied by every PR) | false | 10 test(s) fail on

the PR, e.g.
test.with_dummyserver.test_connectionpool.TestConnectionPool::test_connection[h11] |

pytest-pr-1.log:50906 «ERROR

test/with_dummyserver/test_connectionpool.py::TestConnectionPool::test_connection[h11]»

| - | Change not in the PR description (this run) | unmentioned change | 10 test(s) are new in the PR and fail, e.g.

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_connection[h11] | pytest-pr-1.log:50906

«ERROR
test/with_dummyserver/test_connectionpool.py::TestConnectionPool::test_connection[h11]»

| - | Change not in the PR description (this run) | unmentioned change | 2 test(s) ran on main but are missing on

the PR side, e.g.
test.with_dummyserver.test_connectionpool.TestConnectionPool::test_http2_test_error[h11] |

pytest-main-1.log:50852 «130 failed, 340 passed, 18 skipped, 8 xfailed, 145 warnings, 499 errors in 173.94s

(0:02:53)»

Citation check (PASS): 4 citation(s), every one re-read from its log file and matched.

What changed in behavior

| What | Count | Examples

| Tests that actually ran (main / PR, fewest of the runs) | 839 / 865 | | Tests unchanged | 863 | -

| New tests that pass | 18 | | Tests that pass on main but fail on the PR | 0 | | Tests that are new in the PR and fail | 10 |

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_connection[h11],

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_connection[h2],

test.with_dummyserver.test_https.TestHTTPS_TLSv1::test_http2_probe_cached_result_differs[h11],

test.with_dummyserver.test_https.TestHTTPS_TLSv1::test_http2_probe_cached_result_differs[h2],

test.with_dummyserver.test_https.TestHTTPS_TLSv1_1::test_http2_probe_cached_result_differs[h11]

| Tests that ran on main but are missing on the PR side | 2 |

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_http2_test_error[h11],

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_http2_test_error[h2]

| Tests that changed result in another way (for example passed -> skipped) | 0 | | Tests that gave different results between the 3 runs of the same side (flaky) | 0 | | Tests that fail on main and pass on the PR | 0 | | Tests that are new or changed but did not run because an optional module is missing | 0 | | Tests that were re-parametrized (same test, new parameter ids; not counted as removed) | 0 | | Test modules that fail to load on the PR side only (their tests no longer run) | 0 | | Test modules that fail to load on main only (no baseline for their tests) | 0 | | Test modules that fail to load on both sides (their tests did not run) | 0 | | Tests that fail identically on both sides (already broken on main) | 497 |

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_303_redirect_makes_request_lose_body,

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_303_redirect_makes_request_lose_body

_framing[header],
test.with_dummyserver.test_connectionpool.TestConnectionPool::test_303_redirect_makes_r

equest_lose_body_framing[kwarg],

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_bad_connect,

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_bad_decode

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_connection[h11] fails with: failed on setup

with "OSError('failed to bind socket') [single exception in ExceptionGroup]"

test.with_dummyserver.test_connectionpool.TestConnectionPool::test_connection[h2] fails with: failed on setup

with "OSError('failed to bind socket') [single exception in ExceptionGroup]"

test.with_dummyserver.test_https.TestHTTPS_TLSv1::test_http2_probe_cached_result_differs[h11] fails with:

failed on setup with "OSError('failed to bind socket') [single exception in ExceptionGroup]"

Probes

Model-written probes: model provider: nebius (Token Factory, live).

A probe is a small deterministic script run on both sides in the same environments as the tests, 3 alternating runs

per side. It prints JSON observations; plain code (poc/diff.py) compares them. A difference that no claim of the

PR covers needs attention. Every probe was frozen (content sha256) before any PR-side code ran, and checked

against that freeze before each PR-side run. Only a live model probe is blind; fixed probes and recorded model

fixtures are known-answer cases.

diff-api (diff (auto-derived from base-side code, model-free); informational under baseline full,

diff-probes/diff-api.py, frozen e40e9e28cb5a at 2026-11-11T06:19:49.440432+00:00): Public functions touched

by the PR, called with type-appropriate inputs on both sides.

| Observation | Claim | Result | What the probe saw (main -> PR) | Log lines (main, PR)

| urllib3.http2.extract_from_urllib3.import | - | differs, not in the PR description | value: "<absent>" -> {"raises":

"AttributeError"} | no matching log line
probe-diff-api-pr-1.jsonl:1 «"observation":

"urllib3.http2.extract_from_urllib3.import"»

| urllib3.http2.extract_from_urllib3.input0 | - | differs, not in the PR description | value: null -> "<absent>" |

probe-diff-api-main-1.jsonl:1 «"observation": "urllib3.http2.extract_from_urllib3.input0"»
no matching log line

| urllib3.http2.inject_into_urllib3.import | - | differs, not in the PR description | value: "<absent>" -> {"raises":

"AttributeError"} | no matching log line
probe-diff-api-pr-1.jsonl:2 «"observation":

"urllib3.http2.inject_into_urllib3.import"»

| urllib3.http2.inject_into_urllib3.input0 | - | differs, not in the PR description | value: null -> "<absent>" |

probe-diff-api-main-1.jsonl:2 «"observation": "urllib3.http2.inject_into_urllib3.input0"»
no matching log line

| urllib3.util.ssl_.ssl_wrap_socket.input0 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:3 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input0"»
probe-diff-api-pr-1.jsonl:3 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input0"»

| urllib3.util.ssl_.ssl_wrap_socket.input1 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:4 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input1"»
probe-diff-api-pr-1.jsonl:4 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input1"»

| urllib3.util.ssl_.ssl_wrap_socket.input10 | - | probe could not run | same on both sides: {"raises": "AttributeError"}

| probe-diff-api-main-1.jsonl:5 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input10"»
probe-diff-api-pr-1.jsonl:5 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input10"»

| urllib3.util.ssl_.ssl_wrap_socket.input11 | - | probe could not run | same on both sides: {"raises": "AttributeError"}

| probe-diff-api-main-1.jsonl:6 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input11"»
probe-diff-api-pr-1.jsonl:6 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input11"»

| urllib3.util.ssl_.ssl_wrap_socket.input2 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:7 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input2"»
probe-diff-api-pr-1.jsonl:7 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input2"»

| urllib3.util.ssl_.ssl_wrap_socket.input3 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:8 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input3"»
probe-diff-api-pr-1.jsonl:8 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input3"»

| urllib3.util.ssl_.ssl_wrap_socket.input4 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:9 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input4"»
probe-diff-api-pr-1.jsonl:9 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input4"»

| urllib3.util.ssl_.ssl_wrap_socket.input5 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:10 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input5"»
probe-diff-api-pr-1.jsonl:10 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input5"»

| urllib3.util.ssl_.ssl_wrap_socket.input6 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:11 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input6"»
probe-diff-api-pr-1.jsonl:11 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input6"»

| urllib3.util.ssl_.ssl_wrap_socket.input7 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:12 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input7"»
probe-diff-api-pr-1.jsonl:12 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input7"»

| urllib3.util.ssl_.ssl_wrap_socket.input8 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:13 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input8"»
probe-diff-api-pr-1.jsonl:13 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input8"»

| urllib3.util.ssl_.ssl_wrap_socket.input9 | - | probe could not run | same on both sides: {"raises": "AttributeError"} |

probe-diff-api-main-1.jsonl:14 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input9"»
probe-diff-api-pr-1.jsonl:14 «"observation":

"urllib3.util.ssl_.ssl_wrap_socket.input9"»

Citation check (PASS): 28 citation(s), every one re-read from its log file and matched.

Differences no claim covers

These differences come from differential probes. No claim in the pull request covers them. The list is for reading

the report. It does not change the verdict.

| Probe | Observation | What the probe saw (main -> PR)

| diff-api | urllib3.http2.extract_from_urllib3.import | value: "<absent>" -> {"raises": "AttributeError"}

| diff-api | urllib3.http2.extract_from_urllib3.input0 | value: null -> "<absent>"

| diff-api | urllib3.http2.inject_into_urllib3.import | value: "<absent>" -> {"raises": "AttributeError"}

| diff-api | urllib3.http2.inject_into_urllib3.input0 | value: null -> "<absent>"

Was the changed code exercised?

Coverage: MEASURED by coverage.py on the PR side (180.9 s run).

Coverage gate (python, confidence normal): 393 of 440 changed executable line(s) ran (89%).

| Changed file | Changed executable lines | Of those, ran

| dummyserver/testcase.py | 2 | 1

| src/urllib3/_base_connection.py | 0 | 0

| src/urllib3/_request_methods.py | 1 | 0

| src/urllib3/connection.py | 160 | 137

| src/urllib3/connectionpool.py | 1 | 1

| src/urllib3/contrib/emscripten/connection.py | 10 | 0

| src/urllib3/contrib/pyopenssl.py | 0 | 0

| src/urllib3/exceptions.py | 1 | 1

| src/urllib3/http2/__init__.py | 0 | 0

| src/urllib3/http2/connection.py | 260 | 248

| src/urllib3/poolmanager.py | 2 | 2

| src/urllib3/util/ssl_.py | 3 | 3

How this is measured: one extra, untimed run of the PR side's tests under coverage.py. Changed lines come

from comparing the tested main and PR files. Compiled code (C, C++, Cython) is invisible to coverage.py; such a

change only counts as verified when the PR adds or changes a test for that module.

What went right

• Tested 2561b91ee0c1 of the PR against main (source-pair route).

• 3 alternating runs per side: main 3, PR 3 of 3.

• pytest exit codes explained: pytest exit 1; 497 failure(s)/error(s) pre-existing and identical on both sides.

• Changed code was exercised (coverage gate): 393 of 440 changed executable line(s) ran (89%).

• Probes ran on both sides: 1 probe(s), 3 alternating runs per side, every run printed its observations.

• Single-thread environment: OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0.

• Machine type recorded: local x86_64 8 cpus.

• Nothing posted upstream: GitHub was only read.

• Known-answer replay (fresh re-run): replayed 2026-10-10 06:30:21 IST, harness 1.0.

• Every claim cites a real log line: 4 citation(s), every one re-read from its log file and matched.

• Every probe observation cites its output line: 28 citation(s), every one re-read from its log file and matched.

What went wrong and how to fix it

| Issue | Status | Detail | How to fix

| Behavior difference found | finding | 10 new test(s) fail on the PR; 2 test(s) removed on the PR side; 839 test(s)

ran on main, 865 on the PR; 18 new test(s) pass, 366 unchanged (26 of them skipped on both sides); 497 test

failure(s) pre-existing and identical on both sides | This is about the PR, not the run: read the failing tests and

their log lines and the Probes section, then decide.

Evals and guardrails

| Check | Result | Detail

| 3 alternating runs per side | PASS | main 3, PR 3 of 3

| pytest exit codes explained | PASS | pytest exit 1; 497 failure(s)/error(s) pre-existing and identical on both sides

| Changed code was exercised (coverage gate) | PASS | 393 of 440 changed executable line(s) ran (89%)

| Probes ran on both sides | PASS | 1 probe(s), 3 alternating runs per side, every run printed its observations

| Single-thread environment | PASS | OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0

| Machine type recorded | PASS | local x86_64 8 cpus

| Nothing posted upstream | PASS | GitHub was only read

| Known-answer replay (fresh re-run) | PASS | replayed 2026-10-10 06:30:21 IST, harness 1.0

| Every claim cites a real log line | PASS | 4 citation(s), every one re-read from its log file and matched

| Every probe observation cites its output line | PASS | 28 citation(s), every one re-read from its log file and

matched

Checklist

- I read the verdict and the plain-English summary.
- Every 'false' or 'unmentioned change' row has a log line I can open.
- Nothing under 'What went wrong' is a run problem (those mean Re-run, not Reject).
- I recorded my decision (see below).

Decision recorded: none yet. To record one, run make decide

REPORT=poc-main/reports/daily/2026-11-11/urllib3-5278 DECISION=approve (or reject,

approve_with_notes, re_run).

Rerun command

make check PR=https://github.com/urllib3/urllib3/pull/5278

Evidence links

- Pull request urllib3/urllib3#5278
- record.json
- pr.diff
- setup-main.log
- setup-pr.log
- git.log
- pytest-main-1.log
- pytest-pr-1.log
- pytest-main-2.log
- pytest-pr-2.log
- pytest-main-3.log
- pytest-pr-3.log
- junit-main-1.xml
- junit-pr-1.xml

• coverage-pytest.log

• coverage-pr.json

• probe-diff-api-main-1.jsonl

• probe-diff-api-pr-1.jsonl

• main run 1 result: pytest-main-1.log:50852 «130 failed, 340 passed, 18 skipped, 8 xfailed, 145 warnings, 499

errors in 173.94s (0:02:53)»

• pr run 1 result: pytest-pr-1.log:51413 «130 failed, 358 passed, 18 skipped, 8 xfailed, 145 warnings, 507 errors in

173.93s (0:02:53)»

Glossary

| Term | Meaning

| main | The library as it is today, without the PR.

| PR | Pull request: a proposed change to the library.

| run | One full pass of the selected tests. We do 3 per side, alternating main and PR.

| flaky | A test that gives different results when nothing changed; it is ignored as evidence.

| log line | A line in a pytest-*.log file: the raw output the verdict is based on.

| route | How the two sides were built: nightly+patch, source-build or source-pair (see docs/OPERATIONS.md).

| changed lines covered | How many of the PR's changed, executable Python lines ran during the PR side's tests.

| re-parametrized | A test whose parameter ids changed (renamed cases); the new ids are judged on their own.

rebuilt from cleaned text

Written 2026-11-11 12:44 IST